Answer Key — Why the Moon Has Phases (and Locating It)

Final answer with a one-line reason. Full methods are in the Notes' worked examples.

Objective

Q1 **(b) the changing Sun–Moon–Earth positions** — phases come from the changing relative positions, not Earth's shadow.

Q2 **(b) full Moon day** — a lunar eclipse happens at full Moon (Earth's shadow on Moon).

Q3 **(b) closest** — at new Moon the Moon is closest to the Sun in the sky.

Q4 **(b) 50 minutes later** — the Moon rises about 50 minutes later each day.

Q5 **(a) Both true, R explains A** — the Moon between Earth and Sun (new Moon) causes a solar eclipse — as R explains.

Q6 **lunar** — Earth's shadow on the Moon is a lunar eclipse.

Q7 **sunset** — a waxing Moon is best seen at sunset.

Short answer

Q8 The Sun lights one half of the Moon; as the Moon revolves around Earth, we see different fractions of that lit half, so its bright shape appears to change — the phases.

Q9 Lunar eclipse: Earth's shadow falls on the Moon, on a full Moon day. Solar eclipse: the Moon comes between Earth and Sun, on a new Moon day.

Q10 Because the Moon rises about 50 minutes later each day, on many days it rises during the afternoon and is above the horizon while the Sun is still up.

Long / application

Q11 At new Moon the Moon lies roughly between Earth and Sun, so it is near the Sun in the sky and its lit half faces away from us (not visible). At full Moon the Earth is between Sun and Moon, so the Moon is opposite the Sun (rises at sunset) and its whole lit half faces us (full circle). The position relative to the Sun decides how much lit Moon we see.

Q12 Because the Moon's orbit is slightly tilted relative to Earth's orbit around the Sun. Most months the Sun, Earth and Moon don't line up exactly, so the shadow misses, and no eclipse occurs — eclipses happen only when the alignment is right.

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